Neel Gali

ML Systems Engineer

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PROFESSIONAL SUMMARY

ML Systems Engineer with 3+ years of experience in High Performance Computing and building scalable systems for deep learning model training and prediction. Proficient in C/C++ programming within a Linux environment, with a strong understanding of operating system concepts and computer architecture. Expertise in optimizing mainstream machine learning frameworks like TensorFlow and PyTorch for ML Hardware Acceleration, including GPU/RDMA, to achieve significant performance optimizations through model pruning and quantization.

TECHNICAL SKILLS

Programming & Systems: C/C++ programming, Python, Bash, Linux environment, Operating System Concepts, Computer Architecture, Scalable Systems

Machine Learning & Deep Learning: TensorFlow, PyTorch, Applied ML models (MLP/Transformer/MoE), Deep Learning Model Training, Prediction, Algorithm Implementation, Model Pruning, Quantization, Compression Methods, Scikit-learn, OpenCV, YOLOv8

Performance & Hardware Acceleration: High Performance Computing (HPC), ML Hardware Acceleration, GPU/RDMA, Software and Hardware Integration, Optimizations, CUDA

Cloud & DevOps: AWS, Azure, GCP, Docker, Kubernetes, Git, CI/CD, Terraform

PROFESSIONAL EXPERIENCE

ML Systems Engineer | Epik Solutions

Jun 2025 - Present

• Engineered and deployed high-performance ML inference systems for real-time cybersecurity threat detection, utilizing C/C++ programming and Python within a Linux environment to process terabytes of telemetry data; improved detection latency by 40% and reduced false positives by 25% for security analysts, enhancing proactive threat intelligence.

• Collaborated with security and infrastructure stakeholders to define system requirements and architect scalable systems for Applied ML models, ensuring robust deep learning model training and efficient prediction pipelines; facilitated 30% faster model iteration cycles, directly supporting strategic threat intelligence initiatives.

• Designed and implemented optimized computer vision pipelines using PyTorch and TensorFlow, focusing on ML Hardware Acceleration through GPU/RDMA for processing large-scale sensor data; achieved a 2x speedup in data processing throughput, enabling real-time anomaly detection in operational environments.

• Developed robust data preprocessing and feature engineering modules optimized for High Performance Computing environments, handling petabytes of cybersecurity event logs for downstream ML systems; reduced data preparation time by 50%, significantly accelerating deep learning model training for new threat vectors.

• Implemented advanced algorithm implementation for time-series anomaly detection on complex telemetry data, leveraging underlying framework understanding of custom data structures for efficiency; detected 15% more emerging security threats proactively, enhancing overall system resilience.

• Streamlined deployment and monitoring workflows for ML systems using C/C++ and Bash scripts, integrating with existing Linux environment tools for continuous integration; improved deployment reliability by 35% and reduced troubleshooting time by 20% for operational teams.

• Contributed to software and hardware integration efforts for ML inference engines on custom accelerators, performing rigorous validation and optimization of existing algorithms in production; achieved 1.5x performance gain post-deployment by identifying and resolving critical bottlenecks.

ML Engineer | Samay Systems LLC

July 2024 - Jun 2025

• Designed and trained a suite of Applied ML models, including Transformer-based architectures, for compliance and risk analysis workflows, focusing on efficiency for scalable systems; achieved 92% accuracy in identifying compliance deviations, leading to a 20% reduction in audit preparation time.

• Developed robust Python services for data collection and validation from multi-cloud environments, ensuring consistent, high-quality inputs for deep learning model training; streamlined data ingestion by 45%, significantly improving the throughput of ML systems.

• Implemented ML Systems for AI-assisted compliance workflows by integrating LLMs with enterprise data sources, leveraging a deep understanding of computer architecture for efficient data retrieval; enhanced automated compliance checks by 60%, reducing manual review effort.

• Developed and optimized RAG-style querying mechanisms over large-scale compliance datasets, ensuring efficient data retrieval and contextual answer generation via underlying framework adjustments; increased user query response speed by 30% and improved satisfaction for compliance officers.

• Designed and optimized complex relational schemas and SQL workflows to support multi-tenant, multi-framework compliance validation at scale, underpinning robust ML systems for data integrity; supported a 25% expansion in client base without compromising performance.

• Managed and optimized cloud-native deployments of ML systems on AWS, utilizing Docker and Kubernetes for containerization and orchestration, ensuring high availability and scalable systems; achieved 99.9% uptime for critical compliance services, benefiting thousands of users.

ML Systems Intern | Monarch Tractor

Jun 2023 - Aug 2023

• Supported the development of High Performance Computing pipelines for computer vision and ML models in autonomous agricultural vehicles, focusing on real-time inference within a Linux environment; contributed to a 15% improvement in object detection accuracy for critical safety features.

• Engineered Python-based data preprocessing pipelines to transform raw sensor and image data into optimized datasets, significantly accelerating deep learning model training for detection and prediction tasks; reduced data preparation time by 30% for the ML engineering team.

• Developed and validated object detection and segmentation models using PyTorch and TensorFlow, focusing on ML Hardware Acceleration techniques for deployment on embedded systems; achieved 90% real-time perception accuracy under varied field conditions for autonomous operations.

• Performed rigorous testing and validation of ML models using CLI-driven workflows in a Linux environment, gaining hands-on experience with operational ML systems and their software and hardware integration; identified and resolved 5 critical performance bottlenecks prior to field deployment.

• Collaborated with cross-functional engineering teams to refine requirements for Applied ML models and validate outputs against real-world field expectations, ensuring alignment with overall scalable systems architecture; contributed to the successful deployment of a new perception module.

• Participated in deploying optimized ML models to cloud-hosted production environments, gaining practical experience in performance monitoring and inference validation within large-scale ML systems; contributed to a 10% reduction in inference latency through proactive monitoring.

Data Scientist | ZeMoSo Technologies

Jul 2021 - Aug 2022

• Contributed to the development of a cloud-based scientific operations platform, focusing on building scalable systems for tracing experimental data and ensuring data integrity; supported the tracking of 1000+ experiments, improving data auditability by 50%.

• Developed backend services and ML-adjacent pipelines in C++ and Python to validate, enrich, and structure experimental data, creating consistent inputs for downstream Applied ML models; increased data quality score by 20%, ensuring reliable decision-making for scientific teams.

• Automated dataset preparation, feature generation, and experiment tracking for Proof-of-Concept ML experimentation workflows, leveraging operating system concepts for efficient resource management; reduced time-to-insight by 30%, accelerating initial model development.

• Integrated CI/CD pipelines for ML models, enforcing quality checks and promoting artifacts with strong auditability and security controls, crucial for scalable systems; ensured 100% compliance with governance standards, maintaining system integrity.

EDUCATION

Master’s in Computer Science | Oklahoma State University, Stillwater, OK

Bachelor’s in Computer Science | JNTU Hyderabad, India